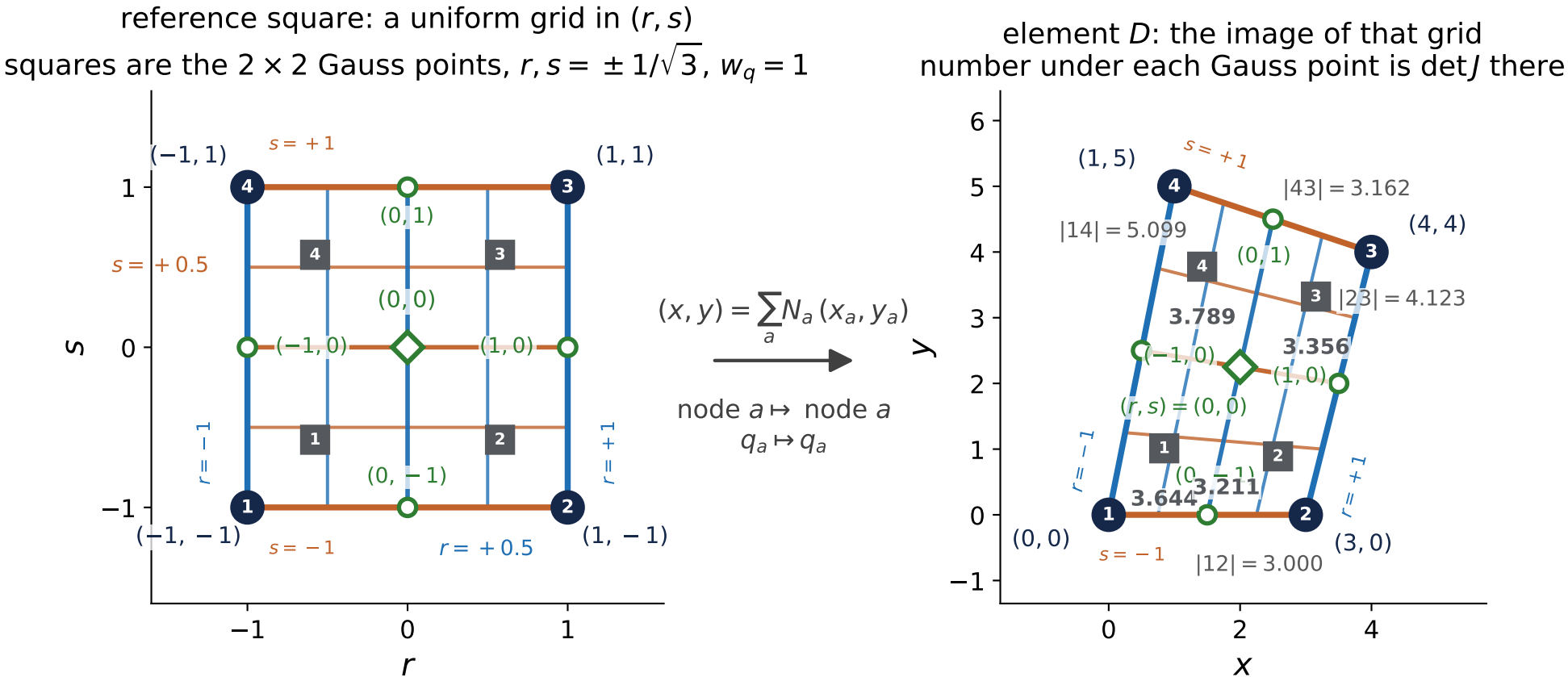

Node a sits at parent corner a, and q_a in node a's corner --- that pairing is the element's numbering, nothing more.
The four edges are not separate from the grid --- they are its outermost members $r = \pm 1$, $s = \pm 1$,
and the middle two are the bimedians, crossing at the image of $(0, 0)$.
Every image line is straight and still evenly divided --- equal steps in r, s stay equal along each line (bottom 0.750, top 0.791).
The distortion is the *fan* between opposite edges, and the cell areas it leaves behind: 0.781 to 0.969, summing to 14.